AISC5020 Assignment 2 for Week 5 to Week 8

Deadline: 3 Nov 2026

Question 1.

Let X and Y be independent random variables sharing the geometric distribution whose mass function is $p(k) = (1-\pi)\pi^k$ for $k = 0,1,2,\ldots$, where $0 < \pi < 1$. Let $U = \min(X,Y)$, $V = \max(X,Y)$, and $W = V - U$. Determine the joint probability mass function for U and W and show that U and W are independent.

Question 2.

Suppose X and Y are conditionally independent random variables having the same Poisson distribution with parameter λ, but where λ is also random, being exponentially distributed with parameter θ. What is the conditional distribution for X given that $X + Y = n$? Note: if X and Y are conditionally independent given $\Lambda = \lambda$, then $\mathrm{P}(X = x, Y = y \mid \Lambda = \lambda) = \mathrm{P}(X = x \mid \Lambda = \lambda)\,\mathrm{P}(Y = y \mid \Lambda = \lambda)$.

Question 3.

Find the mutual information $I(X_1, X_2)$ where X has a bivariate normal distribution:

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim n\left(0, \begin{pmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{pmatrix}\right)$$

Question 4.

Suppose *X, Y* are two points sampled independently and uniformly at random from the interval [0, 1]. What is the expected location of the left most point?

Question 5.

Let a dataset of independent observations be $\mathcal{D} = \{x_1, x_2, \dots, x_N\}$. The empirical distribution is defined as the uniform mixture of Dirac delta measures (or indicator masses):

$$p_{\text{emp}}(x) = \frac{1}{N}\sum_{i=1}^{N} \delta\,(x - x_i)$$

Let $X \sim q(x|\theta)$ be some model with parameter θ. By definition, the maximum likelihood estimate is:

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \sum_{i=1}^{N} \ln q\,(x_i|\theta)$$

Show that $\underset{q}{\operatorname{argmin}}\, D\big(p_{\text{emp}} \big\| q\big)$ is obtained by $q(x) = q\big(x; \hat{\theta}_{\text{MLE}}\big)$.

Question 6.

Let $x \in \{0,1\}$ denote the result of a coin toss: $x = 0$ for tails, $x = 1$ for heads. The coin is potentially biased, so that heads occurs with probability θ_1. Suppose that someone else observes the coin flip and reports to you the outcome, y. But this person is unreliable and only reports the result correctly with probability θ_2; that is $p(y|x, \theta_2)$ is given by

	$y = 0$	$y = 1$
$x = 0$	θ_2	$1 - \theta_2$
$x = 1$	$1 - \theta_2$	θ_2

Assume that θ_2 is independent of x and θ_1.

(a) Write down the joint probability distribution $p(x, y|\theta)$ as a 2×2 table, in terms of $\theta = (\theta_1, \theta_2)$.

(b) Suppose have the following dataset: $x = (1,1,0,1,1,0,0)$, $y = (1,0,0,0,1,0,1)$. What are the MLEs for θ_1 and θ_2? Justify your answer. What is $p\big(Data\big|\hat{\theta}, M_2\big)$ where M_2 denotes this 2-parameter model?

(c) Now consider a model with 4 parameters, $\theta = \big(\theta_{0,0}, \theta_{0,1}, \theta_{1,0}, \theta_{1,1}\big)$, representing $p(x, y|\theta) = \theta_{x,y}$. What is the MLE of θ? What is $p\big(Data\big|\hat{\theta}, M_4\big)$ where M_4 denotes this 4-parameter model?

(d) Suppose we are not sure which model is correct. We compute the leave-one-out cross validated loglikelihood of the 2-parameter model and the 4-parameter model as follows:

$$L(m) = \sum_{i=1}^{n} \log p\left(x_i, y_i \middle| m, \hat{\theta}(Data_{-i})\right)$$

and $\hat{\theta}(Data_{-i})$ denotes the MLE computed on Data excluding row i. Which model will CV pick and why?

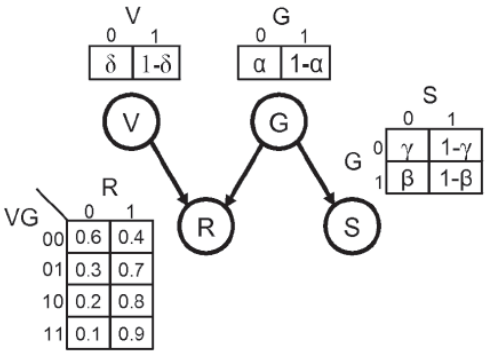
(e) Recall that an alternative to CV is to use the BIC score, defined as

$$BIC(M, Data) = \log p\left(Data \middle| \hat{\theta}_{MLE}\right) - \frac{\text{dof}(M)}{2} \log N$$

where $\text{dof}(M)$ is the number of free parameters in the model. Compute the BIC scores for both models. Which model does BIC prefer?

Question 7.

Consider the DAG in figure 10.15(a):



The DAG consists of 4 binary variables: G = "gray", V = "Vancouver", R = "rain" and S = "sad".

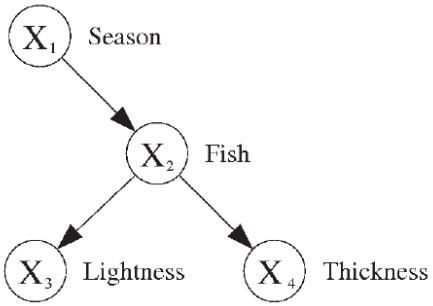
(a) Find $P(S = 1|V = 1)$ in terms of $\alpha, \beta, \gamma, \delta$.

(b) Find $P(S = 1|V = 0)$.

(c) Find MLE of α, β, γ using the following data.

V	G	R	S
1	1	1	1
1	1	0	1
1	0	0	0

Question 8.



Consider the DAG in the above figure. The nodes represent

$$X_1 \in \{winter, spring, summer, autumn\}$$

$$X_2 \in \{salmon, sea\ bass\}$$

$$X_3 \in \{light, medium, dark\}$$

$$X_4 \in \{wide, thin\}$$

The corresponding conditional probability tables are

$$p(x_1) = (0.25 \quad 0.25 \quad 0.25 \quad 0.25), p(x_2|x_1) = \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \\ 0.4 & 0.6 \\ 0.8 & 0.2 \end{pmatrix}$$

$$p(x_3|x_2) = \begin{pmatrix} 0.33 & 0.33 & 0.34 \\ 0.8 & 0.1 & 0.1 \end{pmatrix}, p(x_4|x_2) = \begin{pmatrix} 0.4 & 0.6 \\ 0.95 & 0.05 \end{pmatrix}$$

In each conditional probability matrix $P(A|B)$, rows represent B and columns represent A.

(a) Suppose the fish was caught on December 20 – the end of autumn and the beginning of winter – and thus let $p(x_1) = (0.5 \quad 0 \quad 0 \quad 0.5)$ instead of the above prior. Suppose the lightness has not been measured but it is known that the fish is thin. Classify the fish as salmon or sea bass.

(b) Suppose all we know is that the fish is thin and medium lightness. What season is it now, most likely? Use $p(x_1) = (0.25 \quad 0.25 \quad 0.25 \quad 0.25)$.